EXPERIMENT 14

Introduction to Deluge Scripting: Writing Custom Functions and Basic Programs

Subject: No-Code Application Development

Aim

To understand the fundamentals of Zoho Deluge scripting and design simple expressions and functions used in business application logic.

Objectives

- Recognize variables, conditions and built-in functions.
- Write a basic leave-duration calculation.
- Apply validation and conditional logic in a Zoho Creator workflow.

Selected Application

Employee Leave Management application with Start_Date, End_Date and Status fields. Field link names depend on the Creator application.

Requirements / Tools

- Zoho Creator Deluge editor or approved scripting environment
- Sample dates and fictional request records
- Knowledge of field link names and the workflow trigger

No-Code Connection

Deluge represents the low-code extension of a mostly no-code Creator app. Simple expressions supplement visual form rules when calculation or workflow logic becomes too specific.

Illustrative Business Scenario

Compute inclusive leave duration, reject reversed dates and display a helpful validation message, then compare against built-in no-code rules.

Experimental Procedure

Follow these steps in the designated simulator, browser or visual app builder. Menu locations and available features may vary by product version and account plan.

1. Open a workflow or custom-function editor in Zoho Creator.
2. Create variables to store numeric or text values and use info statements for diagnostics.
3. Add a conditional expression for a positive leave duration.
4. Use the date difference function supported by the configured Deluge environment to estimate days.
5. Test sample dates and verify that missing or reversed dates are handled.
6. Attach the validated logic to a suitable form event and save only after a test run.

No-Code Application Perspective

- Identify the user action and the visual interface element involved (form, page, workflow or report).
- Trace the data movement from input to processing to stored record, report or notification.
- Identify configuration that replaces traditional hand-written code and note where scripting might still be required.
- Check whether the experiment can be repeated using sample records and documented settings.

Evidence to Collect

- Screenshot of the configured screen, topology, tool or application.
- A short observation table showing input, action and output.
- One successful test and one failed or exceptional test with expected behavior.

Deluge concepts

Concept	Illustrative syntax
Variable assignment	days = 3;
Conditional	if(days > 0) { ... }
Output / diagnostics	info days;
String values	status = "Pending";

Example tests

Input	Expected outcome
Start 2026-10-12, End 2026-10-12	1 calendar day inclusive
Start 2026-10-12, End 2026-10-14	3 calendar days inclusive
End earlier than start	Validation error

Interpretation / Analysis

- Deluge supports procedural logic that complements visual workflow builders.
- Date differences are often exclusive of the start date; an inclusive day count adds one to the difference.
- Field names, function signatures and trigger behavior should be verified inside the actual Creator editor.

Evaluation Against a No-Code Solution

- Ease of creation: determine how much is configured visually rather than programmed.
- Data and integration: identify where records, connections or services are created and managed.
- Governance: consider role permissions, traceability and safe data handling.
- Limitations: identify any feature that may require paid access, external services or custom logic.

Verification / Test Cases

Suggested Test Record

Test	Input / action	Expected observation
Normal test	Valid sample values and supported action	Expected successful output
Boundary test	Missing value, invalid input or empty data	Validation or clear empty state
Access test	Unapproved user attempts restricted action	Access limited according to role
Repeatability	Re-run same valid example	Consistent output or explain differences

Observation Notes

During the practical session, replace the illustrative outcomes with actual observed results. Include the software/platform name, version if known, screenshots, and any errors encountered.

Results

The sample function demonstrates variable assignment, input checks and inclusive leave-day calculations, with cases for invalid dates.

Conclusion

Basic Deluge expressions enable tailored field validation and business rules that go beyond standard form settings.

No-Code Learning Outcome

This experiment links the technical concept to a practical no-code application development workflow. It highlights visual configuration, user experience, data handling and safe validation rather than treating the exercise as conventional full-code software development.

Precautions

- Use only fictional sample records for classroom demonstrations; avoid exposing passwords, tokens, employee information or real contact details.
- Record your own screenshots, observed outputs and deviations after executing the experiment. The examples below are models, not verified live results.
- Check access rights before sharing or publishing an app, report, workflow or externally reachable service.